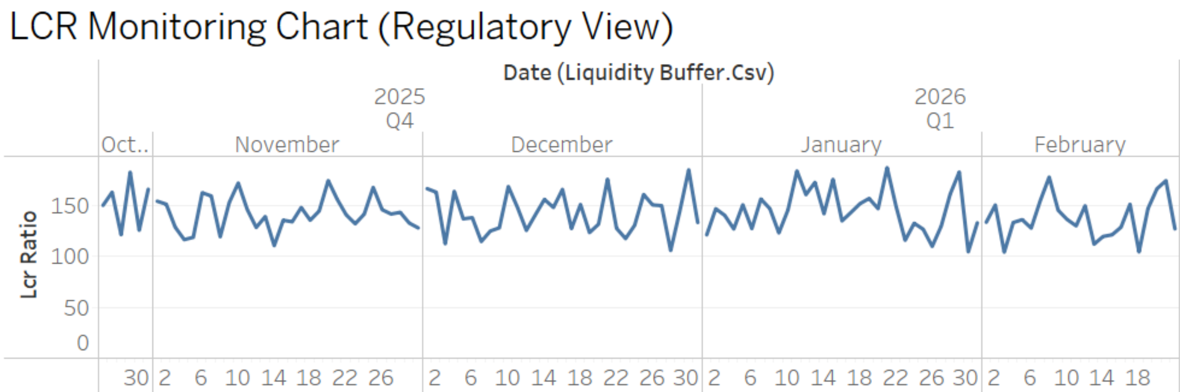
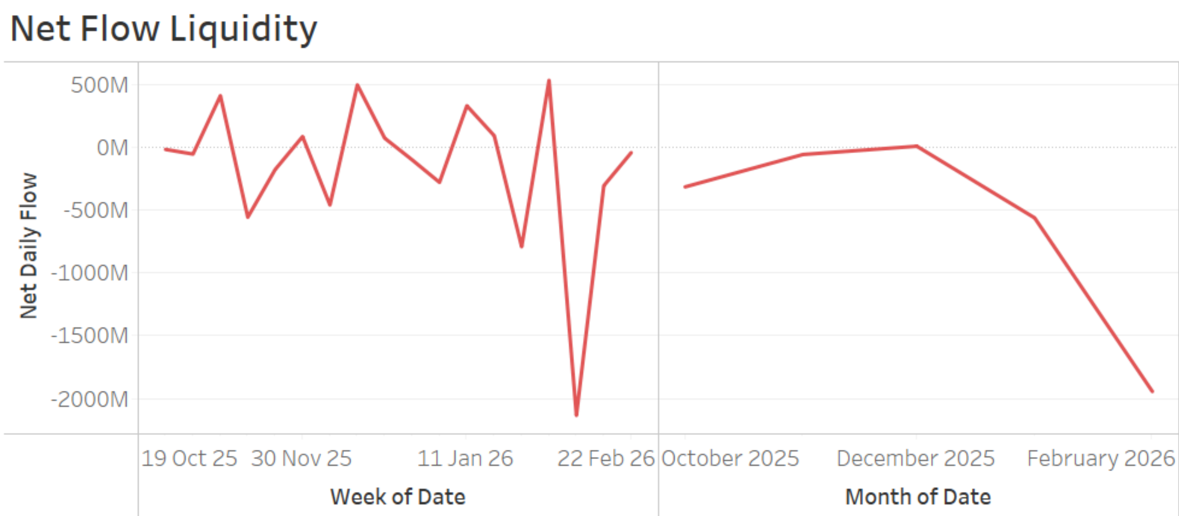
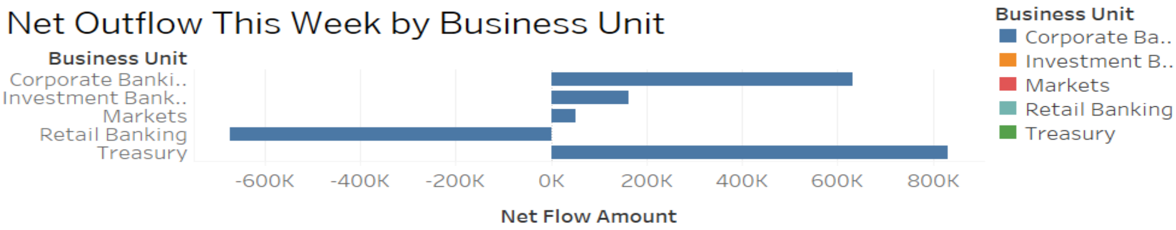
Dashboard Page 1- **Regulatory Liquidity Overview**

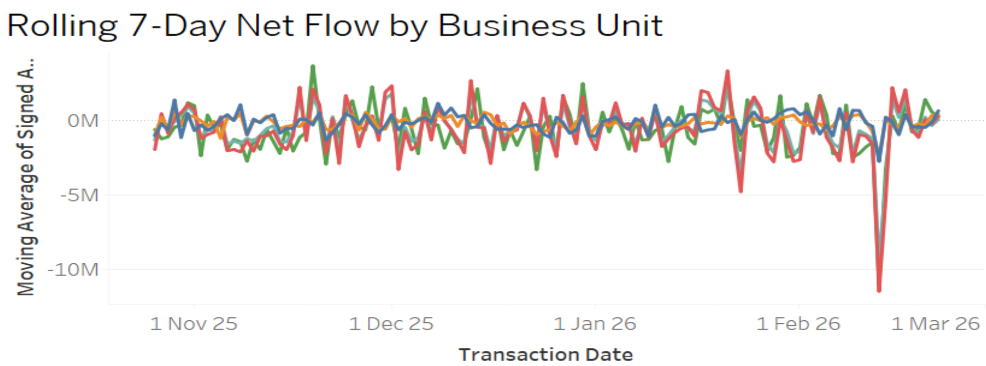



This page provides a high-level view of liquidity stability through the LCR Monitoring Chart and Net Flow Liquidity trends. It highlights regulatory buffer compliance over time and identifies periods of liquidity compression driven by abnormal net outflows. The combined weekly and monthly views help detect stress events and sustained liquidity pressure.

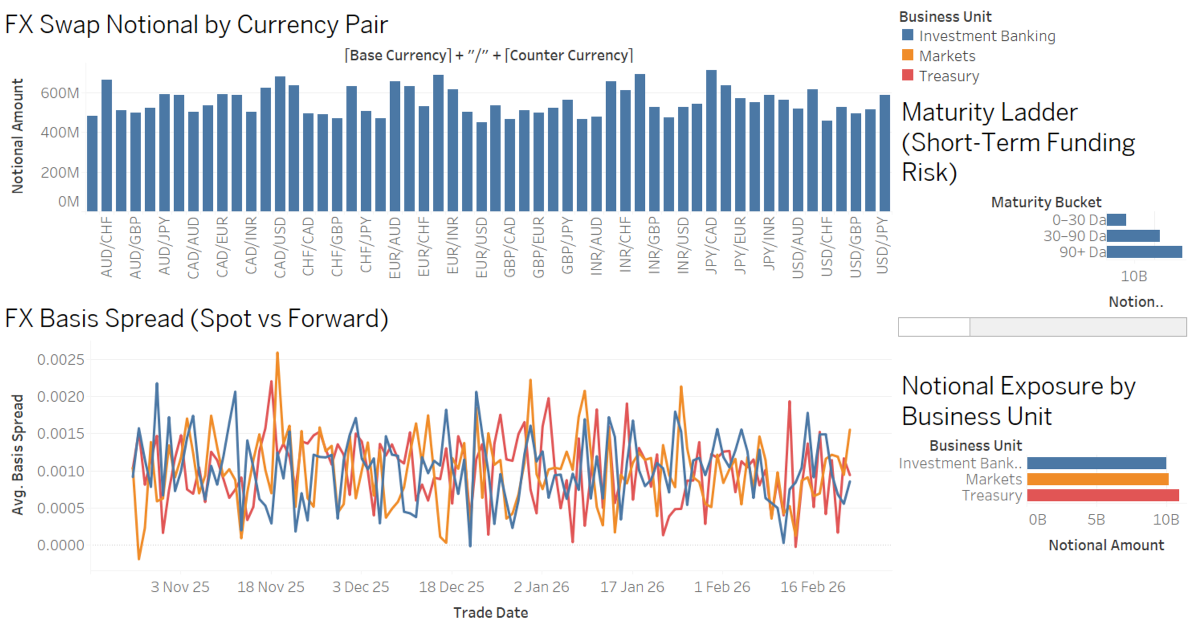
Dashboard Page 2 - **Business Unit Liquidity Contribution**



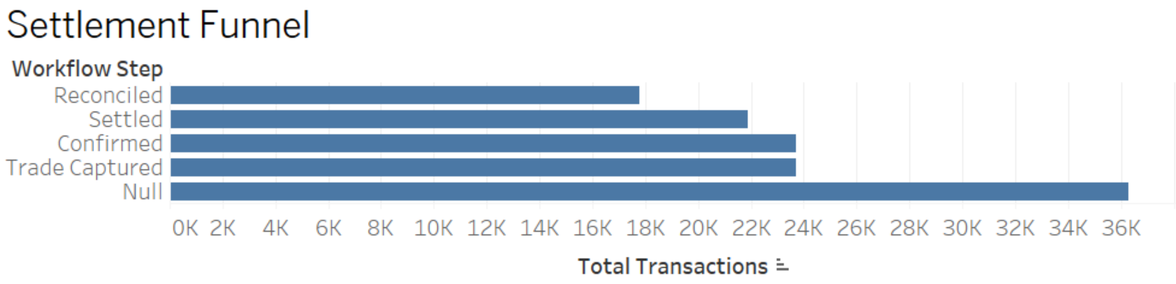

This dashboard analyzes liquidity movements at the business unit level, showing current net outflows and rolling 7-day net flow trends. It helps identify which divisions are contributing most to liquidity volatility and whether recent outflows are short-term fluctuations or sustained pressure. The rolling average view smooths volatility to reveal structural funding patterns.

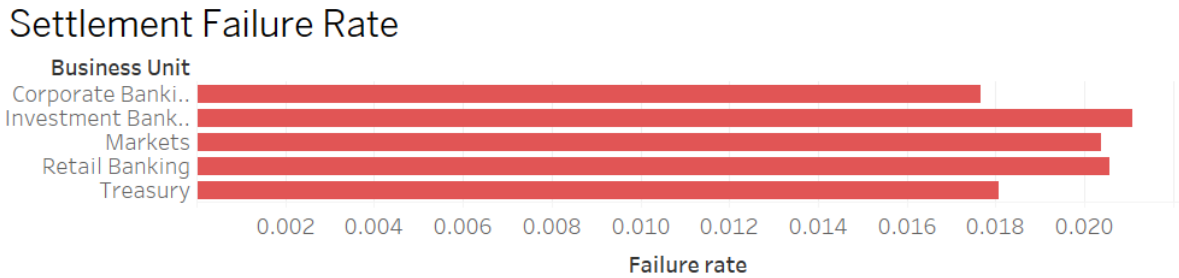
Dashboard Page 3 - **FX Funding & Exposure Risk**




This page focuses on short-term funding risk through FX swap notional exposure, basis spread dynamics, and maturity ladder distribution. It highlights concentration risk by currency pair and business unit while identifying rollover exposure in the 0–30 day bucket. The basis spread chart provides insight into funding cost fluctuations over time.

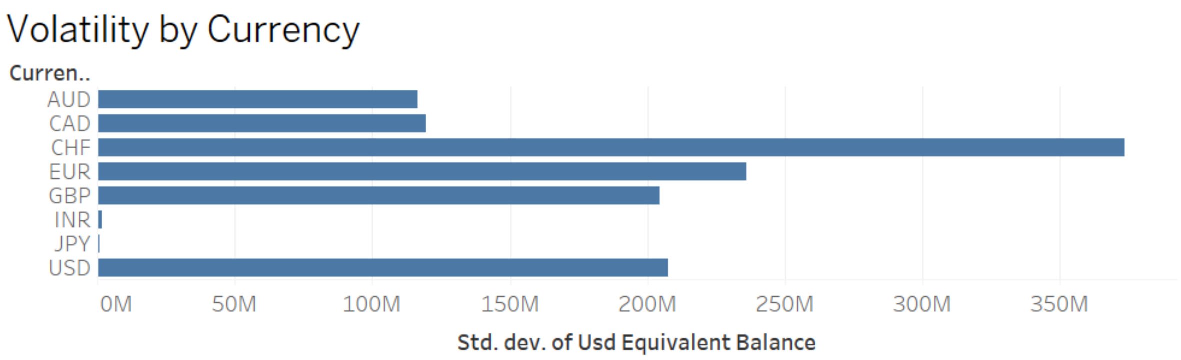
Dashboard Page 4 - **Settlement & Operational Risk**

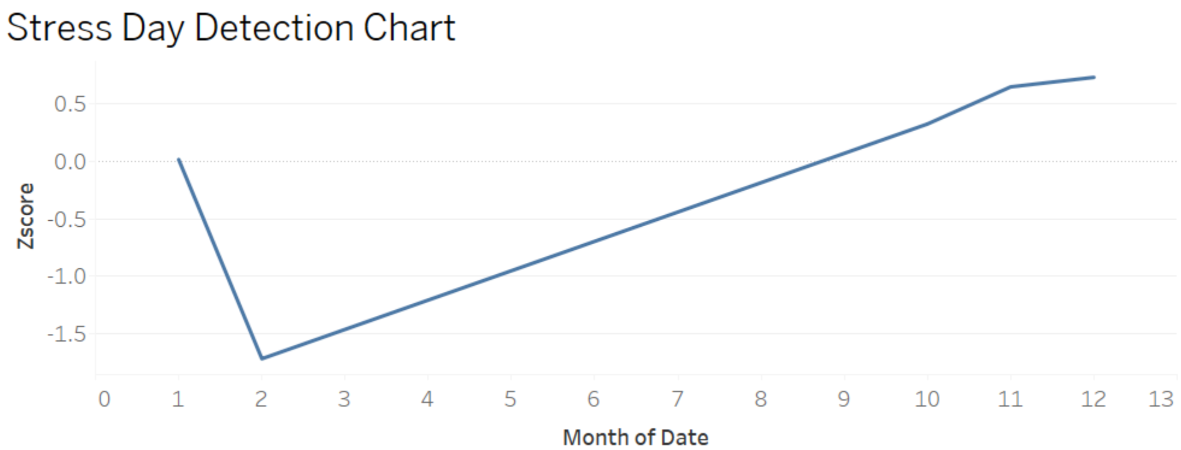



This dashboard evaluates operational efficiency through the settlement funnel and failure rate metrics. It shows workflow drop-offs across settlement stages and identifies business units with elevated failure rates. The analysis links operational delays to potential liquidity timing risk.

Dashboard Page 5 - **Volatility & Stress Detection**




This page measures liquidity volatility by currency using standard deviation of USD-equivalent balances. It also applies statistical stress detection through Z-score analysis to identify abnormal liquidity events. Together, these visuals provide early warning signals for potential liquidity shocks.

All the Pages of the dashboard are dynamic and support cross-filtering.

## Usage of Gen AI:

All the data used in this project was artificially generated using GenAI tools (ChatGPT 5.2). A total of 5 datasets were generated for this project.
ChatGPT was also used to fix errors which occurred while developing visuals / calculation fields.

Steps:

1. Perplexity Pro AI was used to find out information on Bank Of America's business using the job description for "Business Data Analyst" and in general operations of Bank of America.
2. This information was then fed to ChatGPT to generate an artificial dataset for project work.
3. The datasets, although being AI generated, have no personal information and are fully GDPR compliant.

## Information about the Datasets used:

### 1. Cash_transactions.csv

This dataset contains transaction-level cash movements across business units, product lines, and currencies over a 120-day period. Each row represents a single liquidity inflow or outflow event, including settlement status and counterparty type. It is used for net flow analysis, business unit liquidity tracking, and stress day identification.

**Primary Key:** Transaction Id
**Grain:** One row per cash transaction

**Connects to:**

- settlement_workflow_log via **Transaction Id**

### 2. Settlement_workflow_log.csv

This dataset tracks operational settlement workflow steps for each transaction, including timestamps and step sequence. It enables analysis of settlement delays, workflow drop-offs, and operational friction affecting liquidity timing.

**Foreign Key:** Transaction Id
**Grain:** One row per workflow step per transaction

**Connects to:**

- cash_transactions via **Transaction Id**

### 3. Daily_cash_balances.csv

This dataset contains daily opening and closing cash balances for each currency, including net daily flows and FX conversion rates to USD. It supports liquidity trend monitoring, rolling averages, volatility measurement, and FX-adjusted balance analysis.

**Grain:** One row per currency per day

**Connects logically to:**

- liquidity_buffer via **Date**

(Not directly joined to transaction-level data to prevent duplication.)

### 4. Liquidity_buffer.csv

This dataset contains regulatory liquidity metrics, including total High-Quality Liquid Assets (HQLA), net stressed 30-day outflows, and calculated LCR ratio. It is used for regulatory compliance monitoring and stress scenario analysis.

**Grain:** One row per day

**Connects logically to:**

- daily_cash_balances via **Date**

Used for LCR monitoring alongside balance data.

### 5. Fx_swap_positions.csv

This dataset contains FX swap trade-level information, including trade date, maturity date, notional amount, and forward/spot rates. It supports short-term funding analysis, maturity ladder construction, and FX basis spread monitoring.

**Grain:** One row per FX swap trade

**Connects:**

- Operates as an independent funding analysis table
- Shares Business Unit dimension for comparative analysis

It is intentionally not directly joined to transaction or balance tables to avoid data duplication.